Implementation in Cake-cutting problem

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Abstract

This paper formalizes the relationship between fractional cake allocations — partitions of unity — and lotteries over deterministic partitions. It provides an implementation justification for working with fractional allocations: every partition of unity can be implemented by a randomization over feasible deterministic partitions. Moreover, I show that there exists an efficient and envy-free allocation that can be implemented by a lottery with finite support for a broad class of preferences-measures (in particular, atomic).

1. Introduction

2. Literature Review

The famous literature on cake cutting or land division traces back to Steinhaus **?**, who formalized the problem known as the cake-cutting problem, starting the literature on fair division. But this particular definition of fairness, that is, envy-freeness and Pareto efficiency, has not found its way into the literature until Weller **?**. He proved the existence of an EF and PE allocation in the problem of allocating a measurable space $(\Omega, \mathcal{B})$, where agents' preferences are atomless measures on this space. He also showed that such an allocation is supported as a price equilibrium in the exchange economy with equal incomes.

Akin **?** builds on Weller and extends the result to the settings where measures are not necessarily atomless. His notion of envy-freeness is also more general, allowing for assigning non-equal weights to agents' evaluations.

Weller's allocation is a partition, since he assumes atomless measures. Akin's allocation is a partition of unity, since he does not. Both of them give a way to construct such an allocation, though: it is supported as a price equilibrium for certain prices/budgets. My contribution to the literature is that even when measures are not atomless, I show that the allocation in question is a lottery over partitions.

Berliant, Thomson, and Dunz **?** prove the existence of an allocation satisfying PE and the stronger notion of group envy-freeness, but the comestible in the paper is a compact subset of $\mathbb{R}^k$ and agents' utilities are absolutely continuous with respect to Lebesgue measure. They also prove that under certain assumptions, envy-freeness implies group envy-freeness and efficiency. For example, their sufficient conditions are applicable in the problem of allocating sub-intervals of $[0, 1]$ as in Woodall **?**, and the existence of an EF allocation has already been proved.

Hüsseinov and Sagara **?** model a commodity as a measurable space. They relax the convexity assumption of the preferences and prove the existence of a PE and EF allocation

under certain assumptions on the utility function of each individual: uniform continuity, strict monotonicity with respect to set inclusion order, and closed partition matrix range of the utility functions.

3. Set Up and Main Results

The problem of allocating land or, more broadly, allocating a divisible good among agents in a way that satisfies certain fairness and/or efficiency criteria models allocations as partitions of unity.

Definition 3.1. A **partition of unity** on a topological space X is a finite set R of functions from X to the unit interval $[0,1]$ such that for every point $x \in X$ the sum of all the function values at x is 1, i.e.,

$$\sum_{\rho \in R} \rho(x) = 1.$$

Preferences for land (or cake) are modeled as finite measures, and whether it is assumed that the measures are atomless or not calls for conceptually different interpretations of allocations.

The modeling of allocations as partitions of unity is justified as an intermediate technical step. When agents preferences are assumed to be atomless, it is shown that fair allocation can be taken to be a partition.

I claim that regardless of whether utility measures are atomless or not, allocating partitions of unity is formally justified, as they are utility-equivalent to the set the marginal distributions generated by probability measures on the space of partitions. In the following paragraphs, I show that for every probability measure on a deterministic consumption space of land partitions, there exists a utility-equivalent partition of unity, and for every partition of unity there is a utility-equivalent probability measure.

I assume that L is a separable first countable Hausdorff topological space together with its Borel sigma-algebra $\mathcal{B}$. The set of deterministic consumption Y is identical for all agents. It is the set of all measurable subsets of L, $Y = \mathcal{B}$. The set of deterministic allocations $X \subset Y^N$ contains partitions of L into N subsets: $X = \{(F_1, \ldots, F_N) \;\middle|\; F_i \cap F_j = \emptyset \;\; \forall i, j \in N;\; \bigcup_{i \in N} F_i = L\}$.

In order to talk about probability measures on X, start with (Y, τ') as a topological space. I endow it with a relative topology from the power set 2^L. This topology is constructed by extending the Vietoris (or Fell) topology for closed subsets of the space (see the appendix for the extension, **?**). Denote by $\mathcal{M}$ the set of Borel sets of Y, and take $(y, \mathcal{M})$ to be a measure space.

(X, τ) is a topological space with a subset topology inherited from a product topology of Y^N. Denote by $(X, \mathcal{B}')$ a measure space X together with its Borel sigma-algebra.

A probability measure p on the set of deterministic allocations is a mapping from $\mathcal{B}'$ to $[0,1]$ that satisfies the countable additivity property. It generates a family of marginal probability measures $\{p_j\}_{j \in \{1,\ldots,N\}}$, each on the space of all measurable subsets of L.

4. Main results

4.1 Implementation of partitions of unity

Lemma 4.1. *Let P be a probability measure on the space of all measurable subsets of L, $P \in \Delta(\mathcal{B})$. There exists $f : L \to [0,1]$ such that for all finite measures μ on L*

$$\int_{\mathcal{B}} \mu(F) dP(F) = \int_L f(x) d\mu(x)$$

Proof. Observe that $(L, \mathcal{B}, \mu)$ and $(Y, \mathcal{M}, P)$ are σ-finite measure spaces. Define $g : L \times \mathcal{B} \to [0,1]$ as

$$g(y,F) = \begin{cases} 1, & \text{if } y \in F \\ 0, & \text{otherwise} \end{cases}$$

Note that g is nonnegative and $\mathcal{B} \otimes \mathcal{M}$ - measurable (proof in the appendix). Note that for every F, $\mu(F) = \int_L g(y,F) d\mu(y)$. By Tonelli's theorem,

$$\int_{\mathcal{B}} \mu(F) dP(F) = \int_{\mathcal{B}} \int_L g(y,F) d\mu(y) dP(F) = \int_L \int_{\mathcal{B}} g(y,F) dP(F) d\mu(y) = \int_L f(y) d\mu(y)$$

where $f(y) = \int_{\mathcal{B}} g(y,F) dP(F) = P(\{F \in \mathcal{B} : y \in F\})$. By construction $f(L) \subset [0,1]$. □

Therefore, for an agent with preferences μ_j, for every random allocation, there is a bounded map $f_j : L \to [0,1]$ that gives them the same payoff.

Let P be a probability measure on X. It gives rise to a family of marginal probability measures $\{P_j\}_{j \in J}$.

Lemma 4.2. *Let $P \in \Delta(X)$. Then $\sum_{j \in J} f_j = 1$, where f_j is a mapping from Lemma 4.1 corresponding to P_j.*

Proof. For $x \in L$, let $A_j(x) = \{(F_1, \ldots, F_j, \ldots, F_N) : x \in F_j \text{ and } F_k \cap F_l = \emptyset \ \forall k, l \in J; \bigcup_{k \in J} F_k = L\}$. $A_j(x)$ denotes the set of all partitions, in which point x is allocated to agent j. For every $x \in L$, note that the sets $\{A_j(x)\}_{j \in J}$ are pairwise disjoint, that is, $A_i(x) \cap A_j(x) = \emptyset \ \forall i \neq j$: if $(\tilde{F}_1, \ldots, \tilde{F}_N) \in A_j$, then $x \in \tilde{F}_j$ and $x \notin \tilde{F}_i$, hence $(\tilde{F}_1, \ldots, \tilde{F}_N) \notin A_i$. Also for every x, $\cup_{j \in J} A_j = X$. The direction $\cup_{j \in J} A_j \subset X$ is trivial. For the other direction, take $(\tilde{F}_1, \ldots, \tilde{F}_N) \in X$. Since it is a partition, let j denote the set such that $x \in \tilde{F}_j$. Then $(\tilde{F}_1, \ldots, \tilde{F}_N) \in A_j$.

Hence, for every x, $1 = P(X) = P(\bigcup_{j \in J} A_j) = \sum_{j \in J} P(A_j) = \sum_{j \in J} P_j(\{F : x \in F\}) = \sum_{j \in J} f_j(x)$. □

Now that I have established that the set of probability measures on the set of partitions can be embedded into the set of partitions of unity, I prove the other direction: for every partition of unity, there is a lottery on the set of partitions that is utility-equivalent for any family of utilities.

Lemma 4.3. *Let $\{f_j\}_{j \in J}$ be a family of measurable functions $f_j : L \to [0,1]$ such that $\sum_{j \in J} f_j = 1$. Then there exists $P \in \Delta(X)$ such that for every $j \in J$ and every finite measure*

μ on L,

$$\int_X \mu(F_j)\, dP(F_1, \ldots, F_N) = \int_L f_j(x)\, d\mu(x).$$

Proof. Let $J = \{1, \ldots, N\}$ and define $S_0(x) = 0$ and $S_j(x) = \sum_{k=1}^{j} f_k(x)$. For every $\omega \in [0, 1)$, let

$$F_j^\omega = \{x \in L : S_{j-1}(x) \leq \omega < S_j(x)\}$$

and define $\Phi(\omega) = (F_1^\omega, \ldots, F_N^\omega)$. Since S_{j-1} and S_j are measurable, every F_j^ω is measurable. Moreover, for every $x \in L$, the intervals $[S_{j-1}(x), S_j(x))$, $j \in J$, partition $[0, 1)$; hence $(F_1^\omega, \ldots, F_N^\omega)$ is a partition of L for every ω, so $\Phi(\omega) \in X$. Let λ be the Lebesgue probability measure on $[0, 1)$ and let $P = \lambda \circ \Phi^{-1}$. For every $x \in L$ and $j \in J$,

$$P(\{(F_1, \ldots, F_N) \in X : x \in F_j\}) = \lambda(\{\omega : S_{j-1}(x) \leq \omega < S_j(x)\}) = S_j(x) - S_{j-1}(x) = f_j(x).$$

Therefore, by Tonelli's theorem,

$$\int_X \mu(F_j)\, dP = \int_L P(\{(F_1, \ldots, F_N) \in X : x \in F_j\})\, d\mu(x) = \int_L f_j(x)\, d\mu(x).$$

□

4.2 Efficient and envy-free allocations

From now on, the set of stochastic allocations is taken to be the set of partitions of unity on L, instead of the set of probability measures on the set of deterministic allocations. I denote the set of stochastic allocations by Z:

$$Z = \{(f_1, \ldots, f_N) \text{ such that for every } j, f_j : L \to [0, 1] \text{ and } \sum_{j \in J} f_j = 1\}.$$

Following the cake-cutting literature, I embed partitions of unity into $L^\infty(L, \lambda)$ for some measure λ. Usually it is defined as an "average taste" measure $\lambda = \sum_{j \in J} m_j \mu_j$ for a collection of $\{m_j\} \in \mathbb{R}^N$ such that $\sum_{j \in J} m_j = 1, m_j \geq 0$ (agents' weights).

In particular, Berliant **?** takes L to be a compact subset of $\mathbb{R}^2$. He models preferences as measures that are absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^2$. For agent j, the utility from an allocation $(f_1, \ldots, f_N)$ is equal to $U_j(f_j) = \int_L f_j u_j d\lambda$, where λ is the Lebesgue measure and u_j is a corresponding Radon-Nikodym derivative.

The topology on $L^\infty(L, \lambda)$ is a weak-* topology. Observe that the set of allocations Z is closed in the product topology:

$$Z = \bigcap_{j \in 1, \ldots, N} \{(f_1, \ldots, f_N) : f_j \geq 0\} \bigcap \{(f_1, \ldots, f_N) : \sum_{j \in 1, \ldots, N} f_j = 1\}$$

For every j, $\{(f_1, \ldots, f_N) : f_j \geq 0\} = \bigcap_{x \in L} \{(f_1, \ldots, f_N) : f_j(x) \geq 0\}$, where for every $x \in L$, $\{(f_1, \ldots, f_N) : f_j(x) \geq 0\}$ is the preimage of a closed set $[0, \infty)$ under the composition of a linear functional $f_j \to f_j(x)$ with the projection mapping π_j, both continuous by construction.

Analogously, $\{(f_1,\dots,f_N) : \sum_{j\in 1,\dots,N} f_j = 1\} = \bigcap_{x\in L}\{(f_1,\dots,f_N) : \sum_{j\in 1,\dots,N} f_j(x) = 1\}$, and for every x, $\{(f_1,\dots,f_N) : \sum_{j\in 1,\dots,N} f_j(x) = 1\}$ is the preimage of a closed set under the composition $(f_1,\dots,f_N) \in (L^\infty)^N \to \sum f_j \in L^\infty \to \sum f_j(x)$. The summation map is continuous since L^∞ is a TVS.

The last bit is to show that the map $f = (f_1,\dots,f_N) \to (U_1(f),\dots U_N(f))$ is continuous in the weak-* topology of $L^\infty(L,\lambda)$. The proof is due to Akin **?**. Observe that $U_j : f_j \to \int f_j d\mu_j$ is continuous in weak-* topology of $L^\infty(L,\mu_j)$ since $U_j(f_j) = \int f_j g d\mu_j$ with $g \in L^1(\mu_j), g = 1$. Given that μ_j is absolutely continuous wrt μ, $U_j(f_j) = \int f_j g d\mu$ with $g \in L^1(\mu), g = \frac{d\mu}{d\nu}$.

Since L is separable, $L^1(L,\lambda)$ is a separable Banach space, so the unit ball of its dual space $L^\infty(L,\lambda)$ is metrizable and compact (Banach - Alaoglu). Z is a compact metric space, which is also convex and satisfies the anonymity property. The set of its so-called marginals f_j is identical for all agents.

4.3 Finite support

There are several well-established ways to prove that there exists an efficient and envy-free allocation of cake $\bar{f}$ among N agents. One can employ the route of [Vakarova, The Power of Sperner's lemma in Fair Division] to establish existence on the set of probability measures directly, which automatically ensures implementability.

Notice that the existence result for the fractional set of allocations is independent of whether preference measures are atomless or not. Moreover, in both cases, the allocation in question can be characterized in greater detail. To this end, I invoke several results from Dvoretzky, Wald, and Wolfowitz **?**, later referred to as DWW.

Before I state the theorems, let me introduce some definitions. Given an allocation $f \in Z$, let $\nu(f;\mu)$ be the $N \times N$ dimensional vector of agents' utilities from their allocation and the allocation of every other agent

$$\nu(f) = \begin{bmatrix} \int_L f_1(x)d\mu_1(x),\dots,\int_L f_1(x)d\mu_N(x), \\ \dots \\ \int_L f_N(x)d\mu_1(x),\dots,\int_L f_N(x)d\mu_N(x) \end{bmatrix}$$

The set of agents' utilities under all feasible stochastic allocations (partitions of unity) is denoted by $V_N(\mu_1,\dots,\mu_N)$. It is the set of all vactors $\nu(f;\mu)$, where $f \in Z$.

The set of agents' utilities under all feasible stochastic allocations that are simple functions is denoted by $V_N^0(\mu_1,\dots,\mu_N)$. It is the set of all vectors $\nu(f^0;\mu)$, where $f^0 = (f_1^0,...,f_N^0)$ such that $f_j^0 = \sum_k a_k \chi_{E_k}$ and $\sum_{j\in J} f_j = 1$. In other words, f_j is a simple (step) function for every j.

Finally, the set of agents' utilities under all feasible deterministic allocations is denoted by $V_N^*(\mu_1,\dots,\mu_N)$. It is the set of all points $\nu(f^*;\mu)$, where $f^* = (f_1^*,...,f_N^*)$ such that $f_j^* = \chi_{E_j}$ and $\cup_{j\in N} E_j = L$.

Theorem 4.4 (Theorem 3 in DDW)**.** *If measures $\mu_1,\dots,\mu_N$ are finite, the simple range $V_N^0(\mu)$ coincides with $V_N(\mu)$: every point in V_N can be represented as $\nu(f^0)$ for some f^0.*

Theorem 4.5 (Theorem 4 in DDW)**.** *If measures $\mu_1,\dots,\mu_N$ are finite and atomless, the*

degenerate range $V_N^*(\mu)$ coincides with the simple range $V_N^0(\mu)$: every point in V_N^0 can be represented as $\nu(f^*)$ for some f^*.

Observe that 4.4 and 4.5 imply that the sets V_N and V_N^* are equal if $\mu_1, \dots, \mu_N$ are finite atomless measures. Recall that $\bar{f} \in Z$, hence $\nu(\bar{f}) \in V_N$. There exists a vector of indicator functions that constitute a partition ($f = (f_1, ..., f_N) \in Z$ such that $f_j = \chi_{E_j}$ and $\cup_{j \in N} E_j = L$), that gives agents the same utilities as $\bar{g}$. Therefore, a feasible deterministic allocation $\chi_{E_1}, \dots, \chi_{E_N}$ is wPE and EF.

Now, let's consider the case when agents' measures $\{\mu_1, \dots, \mu_N\}$ are not atomless. As in the case of allocating finitely many items, there is little hope that an envy-free allocation is deterministic. However, I can show that wPE and EF allocation can be taken to have finite support. Using 4.4 together with the existence theorem **??**, I know that wPE and EF allocation can be taken to be a family of simple functions. In the next theorem, in the spirit of the Birkhoff-von-Neumann theorem, I show that every such family of simple functions can be represented as a convex combination of families of indicator functions.

Theorem 4.6. *Let $\{f_j^0\}_{j \in J}$ be a family of simple functions that is a partition of unity. Then there exist finitely many families of indicator functions $\{f_{k,j}^*\}_{k \in K, j \in J}$ such that $f_{k,j}^* \geq 0, \sum_{j \in J} f_{k,j}^* = 1$ for every k and such that $(f_1^0, \dots, f_N^0)$ is in the convex hull of $\{(f_{1,1}^*, \dots f_{1,N}^*), \dots, (f_{K,1}^*, \dots f_{K,N}^*)\}$:*

$$\begin{bmatrix} f_1^0 \\ f_2^0 \\ \vdots \\ f_N^0 \end{bmatrix} = \sum_{k=1}^{K} a_k \begin{bmatrix} f_{1,k}^* \\ f_{2,k}^* \\ \vdots \\ f_{N,k}^* \end{bmatrix}$$

where $a_k \geq 0, \sum_{k \in K} a_k = 1$.

Proof. Here, I present a nice and intuitive proof for the case of two agents. WLOG, the two simple functions can be defined on the same partition $\{E_i\}_{i \in n}$ of L. Let $f_1^0 = \sum_{i=1}^n \alpha_i \chi_{E_i}$ and $f_2^0 = \sum_{i=1}^n \beta_i \chi_{E_i}$ such that $\alpha_i + \beta_i = 1$. Order the coefficients for agent 1 in a non-decreasing order $\alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_n$. Then

$$f_1^0 = \alpha_1 \chi_L + (\alpha_2 - \alpha_1)\chi_{L \setminus E_1} + (\alpha_3 - \alpha_2)\chi_{L \setminus (E_1 \cup E_2)} + \dots + (\alpha_n - \alpha_{n-1})\chi_{E_n} + (1 - \alpha_n)\chi_\emptyset$$

Now I claim that

$$f_2^0 = \alpha_1 \chi_\emptyset + (\alpha_2 - \alpha_1)\chi_{E_1} + (\alpha_3 - \alpha_2)\chi_{E_1 \cup E_2} + \dots + (\alpha_n - \alpha_{n-1})\chi_{L \setminus E_n} + (1 - \alpha_n)\chi_L$$

Recall that for every $x \in L$, $f_2^0(x) = \beta_i$ for $E_i \ni x$. Using the form of f_2^0 above, observe for $x \in E_i$

$$f_2^0(x) = (\alpha_{i+1} - \alpha_i)\chi_{E_1 \cup \dots \cup E_i}(x) + (\alpha_{i+2} - \alpha_{i+1})\chi_{E_1 \cup \dots \cup E_{i+1}}(x) + \dots + (1 - \alpha_n)\chi_L(x) =$$

$$= 1 - \alpha_i = \beta_i$$

Note that every pair of indicator functions is defined on a partition, and $\chi_{L\setminus(E_1\cup\cdots\cup E_m)} + \chi_{E_1\cup\cdots\cup E_m} = \chi_L$. Trivially, a telescoping sum $\sum_{i=1}^{n+1}(\alpha_i - \alpha_{i-1})$ with $\alpha_0 = 0$ and $\alpha_{n+1} = 1$ is equal to $\alpha_{n+1} = 1$.

The proof of the generalized version of this theorem for an arbitrary finite number of agents is in the appendix. □

A. Appendix

Vietoris topology on $\mathcal{B}$, the set of measurable subsets of L, is defined with a standard subbase. For every non-empty open subset $U \subset L$, define $[U] = \{F \in \mathcal{B} : F \cap U \neq \emptyset\}$, and define $\langle U\rangle = \{F \in \mathcal{B} : F \subset U\}$. The base for this topology is all sets of the form $\langle U, V_1, \ldots, V_n\rangle = \langle U\rangle \cap \bigcap_{i=1}^{n}[V_n]$, where $U, V_1, \ldots V_n$ are non-empty open subsets of L.

Recall that $g : L \times \mathcal{B} \to [0,1]$ is defined as follows

$$g(y,F) = \begin{cases} 1, & \text{if } y \in F \\ 0, & \text{otherwise} \end{cases}$$

Lemma A.1. *$g : L \times \mathcal{B} \to [0,1]$ is measurable with respect to the product Borel sigma algebra on $L \times \mathcal{B}$.*

Proof. $g^{-1}(1) = \{(y,F) : y \in L, F \in \mathcal{B}, y \in F\}$ is measurable, since it is a complement of a measurable set $\{(y,F) : y \in L, F \in \mathcal{B}, y \in F\}^C = \{(y,F) : y \in F^C\}$. Observe

$$\{(y,F) : y \in F^C\} = \bigcap_{n\in\mathbb{N}} \bigcup_{y\in L} N_n^y(y) \times A(y)$$

where $A(y) = \{F \in \mathcal{B} : y \in F^C\}$, and $\{N_n^y\}_{n\in\mathbb{N}}$ is a countable base of open nbhds of y. $A(y)$ is open in Vietoris topology, since $A(y) = \langle L/y\rangle$, where L/y is open since the topology on L is Hausdorff. □

Theorem A.2. *Let $\{f_j^0\}_{j\in J}$ be a family of simple functions such that $f_j \geq 0, \sum_{j\in J} f_j = 1$. Then there exist finitely many families of indicator functions $\{f_{k,j}^*\}_{k\in K, j\in J}$ such that $f_{k,j}^* \geq 0, \sum_{j\in J} f_{k,j}^* = 1$ for every k and such that $(f_1^0, \ldots, f_N^0)$ is in the convex hull of $\{(f_{1,1}^*, \ldots f_{1,N}^*), \ldots, (f_{K,1}^*, \ldots f_{K,N}^*)\}$:*

$$\begin{bmatrix} f_1^0 \\ f_2^0 \\ \vdots \\ f_N^0 \end{bmatrix} = \sum_{k=1}^{K} a_k \begin{bmatrix} f_{1,k}^* \\ f_{2,k}^* \\ \vdots \\ f_{N,k}^* \end{bmatrix}$$

where $a_k \geq 0, \sum_{k\in K} a_k = 1$.

Proof. For any vector of simple functions $f = (f_1, \ldots f_N)$, there is a common partition $E_1, \ldots E_I$ of L such that every f_j is constant on every E_i. I denote values of the functions on each element of the partition by f_j^i.

The proof goes as follows. In the next paragraph, I construct families of indicator functions corresponding to different partitions of L. Then, using Farkas' lemma, I claim there exists a vector of coefficients such that the vector f of simple functions is a convex combination of the chosen vectors of indicator functions.

Recall, N is the number of agents, and I is the number of elements in the partition. For every $m \in \{1, \ldots, \min(I, N)\}$, construct a coarser partition with m elements. How many partitions for a fixed m? Stirling number of the second kind $S(I, m)$. For every partition, choose m agents from N to give this partition, counting each permutation as a different subset of agents. How many permutations? $P(N, m) = \frac{N!}{(N-m)!}$. The claim now is that there exists $\{a_k^m\}$ such that

$$\begin{bmatrix} f_1^0 \\ f_2^0 \\ \vdots \\ f_N^0 \end{bmatrix} = \sum_{m=1}^{\min(I,N)} \sum_{k=1}^{S(I,m)\cdot P(N,m)} a_k^m \begin{bmatrix} \chi_{A_1^{k(m)}} \\ \vdots \\ \chi_{A_j^{k(m)}} \\ \vdots \\ \chi_{A_N^{k(m)}} \end{bmatrix}$$

Let me highlight important characteristics of the sets $(A_1^{k(m)}, \ldots, A_N^{k(m)})$. For every m, k, j, $A_j^{k(m)} = \bigcup E_i$ for some subset of $(1, \ldots, I)$, possibly empty. For every m, for every partition in $S(I, m)$ at most m out of N $A_j^{k(m)}$ are non-empty.

I establish the existence of non-negative coefficients $\{a_k^m\}$ that sum to one using Farkas' lemma. Note that in essence, one needs to prove that a system of linear equations $Pa = f$ admits a non-negative solution, where $a = \{a_k^m\}_{m \in M, k \in K}$, $f = \{f_j^i\}_{j \in J, i \in I}$ and P is a matrix of zeros and ones characterized below. I will also show below that $\sum_{m \in M} \sum_{k \in K} a_k^m = 1$ follows from the partition of unity constraint on f: $\sum_{j \in J} f_j^i = 1 \ \forall i \in I$.

The rows of P correspond to sets E_i and represent how many times a set E_i gets assigned to a particular agent. The index of a row is (j, i), and there are $J \times I$ rows in total. The entry (j, i) has 1 if in a column partition agent j is allocated a set E_i, otherwise it is zero.

The columns represent the assignment of the sets of the original partition $\{E_1, \ldots, E_N\}$ in different partitions. The columns are indexed by $(m, Par, Perm)$, where m represents how coarse a partition is; for every m, Par gives a particular partition with m elements and $Perm$ gives an assignment of m selected agents to the elements of the partition. $(Par, Perm)$ is indexed by $k(m)$. There are $\sum_m S(I, m) \times P(N, m)$ columns.

Let E_i be one of the subsets of the original partition. How many times does the agent j receive this subset in different partitions? Remove the set E_i from the pool of original subsets to be assigned to agents and repeat the procedure for the rest of $E_{k, k \neq i}$. Every row of P features $\sum_m^{min(I-1,N)} S(I-1, m) \cdot P(N, m)$ of 1s.

Every column contains I number of 1s, since every column represents how all $(E_1, \ldots, E_I)$ are allocated in a corresponding partition, corresponding permutation. Fix column $k(m)$, representing partition and permutation $k(m)$. For every block of I rows that corresponds to agent j, for every row i, 1 represents that the set E_i is in a subset that is assigned to agent j.

Hence, if the set E_i in partition $k(m)$ is allocated to agent j, there is 1 in the ith row of the agent j' block of rows, and there are zeros in the ith row of every other agent's block of rows. This description of P contains all the characteristics that are sufficient for the proof of the next claim.

Using Farkas' lemma, if there is no y such that $P^T y \geq 0$ and $f^T y < 0$, then I establish the existence of the coefficients in question.

Lemma A.3. *Let y be such that $P^T y \geq 0$. Then $f^T y \geq 0$.*

Proof. Observe that $f^T y = \sum_{i \in I} \left(\sum_{j \in J} f_j^i y_{ji} \right)$ is the sum of I convex combinations of $\{y_{ji}\}_{j \in J}$. If $f^T y < 0$, then there exists $\tilde{i}$ such that $\sum_{j \in J} f_j^{\tilde{i}} y_{j\tilde{i}} < 0$. WLOG, let's assume $\tilde{i} = 1$. Then there exists a subset of J agents such that $y_{\bar{j}}^1 < 0$ for every such $\bar{j}$. Also WLOG, let's assume agent 1 is in the subset J, so $y_{11} < 0$.

The idea of the proof is to use the fact that $P^T y \geq 0$. For every m, coarser partition and a particular permutation of agents $(m, k(m))$ $\sum_{j,i} P^T_{(m,k(m)),(j,i)} y_{ji} \geq 0$. Take a subset of these inequalities: select an inequality if for $(m, k(m))$, the column $(1,1)$ has 1 in it. I use this subset of inequalities to bound $|y_{11}|$ from above to ultimately show that $\sum_{i \geq 2} \left(\sum_{j \in J} f_j^i y_{ji} \right) \geq |y_{11}|$.

By varying the column $(j,1)$, we can apply the same construction and upper bound to any $y_{\bar{j}1}$ (and $y_{\bar{j}\tilde{i}}$). To see this, fix $\tilde{i}$. For every allocation $(m, k(m))$ in which agent k receives $E_{\tilde{i}}$, there is an allocation that is the same except $E_{\tilde{i}}$ goes to the agent l. Hence, the same $y_{j'i'}$ are picked for the agent l, except $y_{l\tilde{i}}$ is replaced with $y_{k\tilde{i}}$. Choosing $\tilde{i} = 1$ and picking agent $j = 1$ is, in fact, without loss.

Since $P^T y \geq 0$, list all the rows of P^T where the cell $(1,1)$ is equal to 1. The number of selected inequalities is equal to the number of times agent 1 gets E_1, $\sum_m^{min(I-1,N)} S(I,m) \cdot P(N,m)$. Every inequality will have $I-1$ of y_{ji} on the left-hand side. Whenever agent 1 gets E_1, no other agent gets it, so inequalities will not feature $y_{j1}, j \neq 1$.

The goal is to show that

$$\sum_{j \in J} f_j^2 y_j^2 \geq |y_1^1| - \sum_{i \geq 3} \sum_{j \in J} f_j^i y_j^i$$

The procedure is iterative. The main idea is that there are enough inequalities to show first that $\sum_{j \in J} f_j^2 y_j^2 \geq |y_1^1| - c(I)$, where $c(I)$ is some constant. Iteratively proceed to "extract" elements indexed by I from $c(I)$, getting that $\sum_{j \in J} f_j^2 y_j^2 \geq |y_1^1| - c(I-1) - \sum_{j \in J} f_j^I y_j^I$ and so on.

Observe that for every $j \in J$ for the constant $c(I)$ that is independent of j, there exists a partition and a permutation to support

$$y_j^2 \geq |y_1^1| - c(I)$$

There are J inequalities above, and they all hold for any constant $c(I)$ of the type $c(I) = \sum_{i \neq 1,2} y^i_{j(i)}$, $j(i) \in J$ for every i. How many constants $c(I)$ of this type are there? We need to allocate $I-2$ sets to N agents, giving $\sum_{m=1}^{min(I-2,N)} S(I-2,m) \cdot P(N,m)$ constants.

In particular, for every $j \in J$

$$\sum_{j \in J} f_j^2 y_j^2 \geq |y_1^1| - c(I-1) - y_j^I$$

where the inequality holds for any constant of type $c(I-1) = \sum_{i \neq 1,2,I} y^i_{j(i)}$, $j(i) \in J$ for every i. Similarly to the previous paragraph, there are $\sum_{m=1}^{min(I-3,N)} S(I-3,m) \cdot P(N,m)$ constants.

Therefore

$$\sum_{j \in J} f_j^2 y_j^2 \geq |y_1^1| - c(I-1) - \sum_{j \in J} f_j^I y_j^I$$

Iterating, for every $j \in J$

$$\sum_{j \in J} f_j^2 y_j^2 \geq |y_1^1| - c(I-2) - y_j^{I-1} - \sum_{j \in J} f_j^I y_j^I$$

for $c(I-2) = \sum_{i \neq 1,2,I,I-1} y^i_{j(i)}$, $j(i) \in J$ for every i.

Therefore

$$\sum_{j \in J} f_j^2 y_j^2 \geq |y_1^1| - c(I-2) - \left(\sum_{j \in J} f_j^{I-1} y_j^{I-1} + \sum_{j \in J} f_j^I y_j^I \right)$$

Proceeding iteratively, I establish the desired result given that there are enough inequalities to support the claims above.

Lemma A.4. *Let ξ be a natural number such that $\xi < I$. For every such ξ*

$$\sum_{m=1}^{min(I-\xi,N)} S(I-\xi,m) \cdot P(N,m) = N \times \sum_{m=1}^{min(I-(\xi+1),N)} S(I-(\xi+1),m) \cdot P(N,m)$$

Proof. Using the recurrence relation of the Stirling numbers of the second kind,

$$S(I-\xi,m) = mS(I-(\xi+1),m) + S(I-(\xi+1),m-1)$$

I get

$$\sum_{m=1}^{min(I-\xi,N)} S(I-\xi,m) \cdot P(N,m) = \sum_{m=1}^{min(I-\xi,N)} mS(I-\xi,m) \cdot P(N,m) +$$

$$+ \sum_{m=1}^{min(I-\xi,N)} S(I-(\xi+1),m-1) \cdot P(N,m)$$

Note that $S(n,m) = 0$ where $m > n$, so for all $m > I-(\xi+1)$, $S(I-(\xi+1),m) = 0$. Since N is a natural number, $\sum_{m=1}^{min(I-\xi,N)} mS(I-(\xi+1),m) \cdot P(N,m) = \sum_{m=1}^{min(I-(\xi+1),N)} mS(I-(\xi+1),m) \cdot P(N,m)$. For the second term in the sum on the right-hand side, re-labeling produces $\sum_{m=0}^{min(I-\xi,N)-1} S(I-(\xi+1),m) \cdot P(N,m+1)$. $S(n,0) = 0$ for any $n > 0$, and

$P(N, m+1) = (N-m)P(N, m)$. Summing the two terms again produces

$$\sum_{m=1}^{min(I-(\xi+1),N)} mS(I-(\xi+1), m) \cdot P(N, m) + \sum_{m=1}^{min(I-\xi,N)-1} S(I-(\xi+1), m) \cdot P(N, m)(N-m) =$$

$$= N \times \sum_{m=1}^{min(I-(\xi+1),N)} S(I-(\xi+1), m) \cdot P(N, m)$$

since if $N \geq 1-\xi$, $min(1-\xi, N) - 1 = I - (\xi+1)$, and if $N < 1-\xi$, $min(I-(\xi+1), N) = min(I-\xi, N) = N$. □

The claim above establishes the fact that there are just enough inequalities. Every step in the iterative process uses all inequalities from the previous step, and supports the idea used in the construction of the matrix P, that is, allocating subsets of the original partition to agents in all possible ways is necessary to apply Farkas' lemma. □

The last claim to prove is $\sum_{m \in M} \sum_{k(m) \in K(m)} a^m_{k(m)} = 1$. To this end, I show that for any i, $\sum_{m \in M} \sum_{k(m) \in K(m)} a^m_{k(m)} = \sum_{j \in J} f^i_j$. Fix i. By construction $\sum_{k,m(k)} P_{(ji),(m,k(m))} a^m_{k(m)} = f^i_j$. Observe that the 1s in a row (j, i) reflect the partitions in which agent j gets E_i. For every column, every original subset E_i, if agent k gets E_i, then no other agent is allocated E_i. Thus, every $a^m_{k(m)}$ is present at most once in the sum $\sum_{j \in J} \sum_{k,m(k)} P_{(ji),(m,k(m))} a^m_{k(m)}$. On the other hand, since every $m, k(m)$ assigns an agent to all original subsets, including E_i, all $a^m_{k(m)}$ are featured in $\sum_{j \in J} \sum_{k,m(k)} P_{(ji),(m,k(m))} a^m_{k(m)}$. □